

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:

Faculty Name: Prof. S.Gopikrishnan

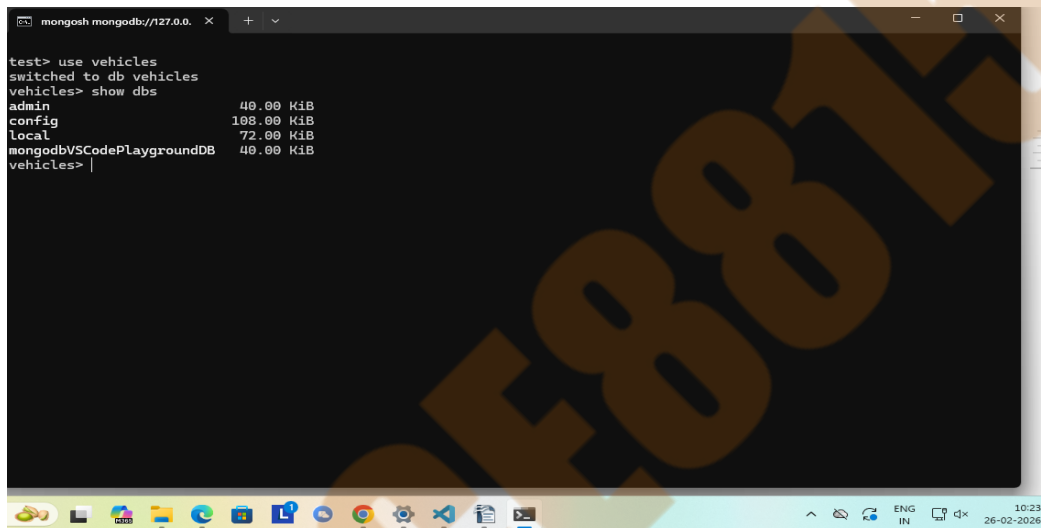
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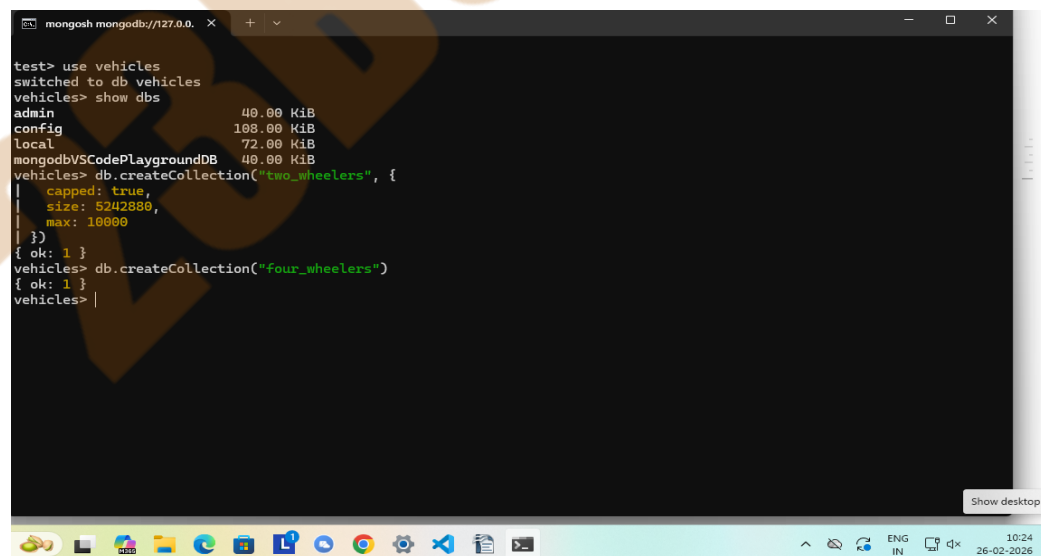
1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".
2. Write a MongoDB query to display all the databases.



```
test> use vehicles
switched to db vehicles
vehicles> show dbs
admin                40.00 KiB
config               108.00 KiB
local                72.00 KiB
mongodbVSCodePlaygroundDB 40.00 KiB
vehicles> |
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



```
test> use vehicles
switched to db vehicles
vehicles> show dbs
admin                40.00 KiB
config               108.00 KiB
local                72.00 KiB
mongodbVSCodePlaygroundDB 40.00 KiB
vehicles> db.createCollection("two_wheelers", {
  capped: true,
  size: 5242880,
  max: 10000
})
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
vehicles> |
```

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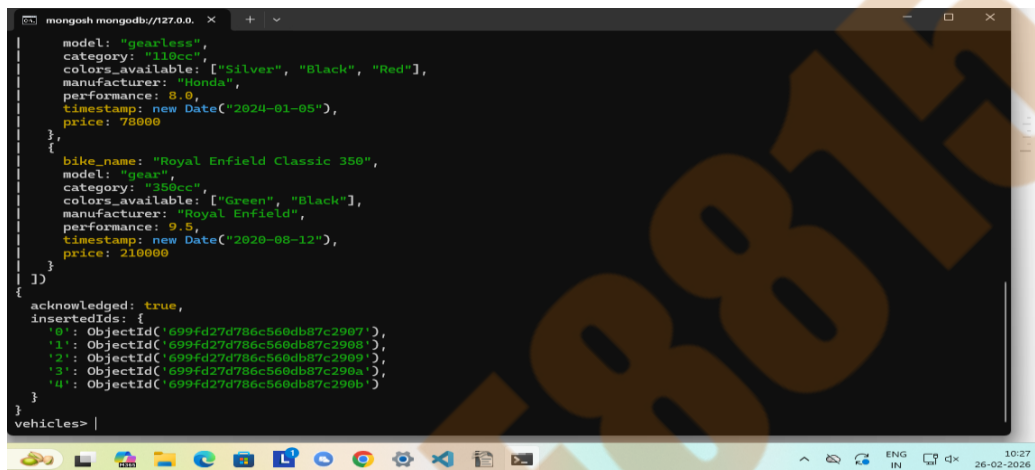
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4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

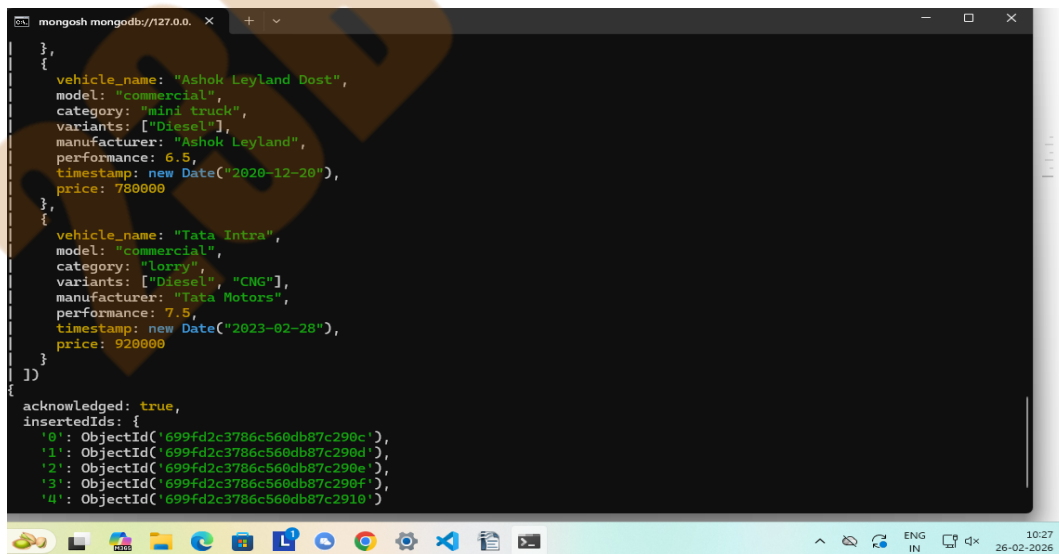


```

model: "gearless",
category: "110cc",
colors_available: ["Silver", "Black", "Red"],
manufacturer: "Honda",
performance: 8.0,
timestamp: new Date("2024-01-05"),
price: 78000
},
{
  bike_name: "Royal Enfield Classic 350",
  model: "gear",
  category: "350cc",
  colors_available: ["Green", "Black"],
  manufacturer: "Royal Enfield",
  performance: 9.5,
  timestamp: new Date("2020-08-12"),
  price: 210000
}
})
acknowledged: true,
insertedIds: {
  '0': ObjectId('699fd27d786c560db87c2907'),
  '1': ObjectId('699fd27d786c560db87c2908'),
  '2': ObjectId('699fd27d786c560db87c2909'),
  '3': ObjectId('699fd27d786c560db87c290a'),
  '4': ObjectId('699fd27d786c560db87c290b')
}
vehicles> |

```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



```

},
{
  vehicle_name: "Ashok Leyland Dost",
  model: "commercial",
  category: "mini truck",
  variants: ["Diesel"],
  manufacturer: "Ashok Leyland",
  performance: 6.5,
  timestamp: new Date("2020-12-20"),
  price: 780000
},
{
  vehicle_name: "Tata Intra",
  model: "commercial",
  category: "lorry",
  variants: ["Diesel", "CNG"],
  manufacturer: "Tata Motors",
  performance: 7.5,
  timestamp: new Date("2023-02-28"),
  price: 920000
}
})
acknowledged: true,
insertedIds: {
  '0': ObjectId('699fd2c3786c560db87c290c'),
  '1': ObjectId('699fd2c3786c560db87c290d'),
  '2': ObjectId('699fd2c3786c560db87c290e'),
  '3': ObjectId('699fd2c3786c560db87c290f'),
  '4': ObjectId('699fd2c3786c560db87c2910')
}

```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

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```
mongosh mongodb://127.0.0.1:27020
vehicles> db.two_wheelers.find().pretty()
[
  {
    _id: ObjectId('699fd27d786c560db87c2907'),
    bike_name: 'Hero Splendor Plus',
    model: 'gearless',
    category: '100cc',
    colors_available: [ 'Red', 'Black', 'Blue' ],
    manufacturer: 'Hero MotoCorp',
    performance: 7.5,
    timestamp: ISODate('2023-03-15T00:00:00.000Z'),
    price: 72000
  },
  {
    _id: ObjectId('699fd27d786c560db87c2908'),
    bike_name: 'Bajaj Pulsar NS200',
    model: 'gear',
    category: '200cc',
    colors_available: [ 'Black', 'Red' ],
    manufacturer: 'Bajaj Auto',
    performance: 9,
    timestamp: ISODate('2022-11-10T00:00:00.000Z'),
    price: 145000
  },
  {
    _id: ObjectId('699fd27d786c560db87c2909'),
    bike_name: 'TVS Apache RTR 160',
    model: 'gear',
    category: '160cc',
    colors_available: [ 'Black', 'Red' ],
    manufacturer: 'TVS Motor Company',
    performance: 8.5,
    timestamp: ISODate('2023-03-15T00:00:00.000Z'),
    price: 85000
  }
]
```

```
mongosh mongodb://127.0.0.1:27020
vehicles> db.four_wheelers.find().pretty()
[
  {
    _id: ObjectId('699fd2c3786c560db87c290c'),
    vehicle_name: 'Maruti Suzuki Swift',
    model: 'own',
    category: 'car',
    variants: [ 'VXi', 'ZXi', 'Petrol', 'Diesel' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8.5,
    timestamp: ISODate('2023-06-10T00:00:00.000Z'),
    price: 650000
  },
  {
    _id: ObjectId('699fd2c3786c560db87c290d'),
    vehicle_name: 'Tata Nexon',
    model: 'own',
    category: 'car',
    variants: [ 'Petrol', 'Diesel', 'EV' ],
    manufacturer: 'Tata Motors',
    performance: 8.8,
    timestamp: ISODate('2022-09-15T00:00:00.000Z'),
    price: 850000
  },
  {
    _id: ObjectId('699fd2c3786c560db87c290e'),
    vehicle_name: 'Mahindra Bolero',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'Petrol', 'Diesel' ],
    manufacturer: 'Mahindra & Mahindra',
    performance: 7.5,
    timestamp: ISODate('2023-06-10T00:00:00.000Z'),
    price: 1200000
  }
]
```

- Write a MongoDB query to display only vehicle name and price in all the collection of the database

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```
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
[
  { bike_name: 'Hero Splendor Plus', price: 72000 },
  { bike_name: 'Bajaj Pulsar NS200', price: 145000 },
  { bike_name: 'TVS Apache RTR 160', price: 125000 },
  { bike_name: 'Honda Activa 6G', price: 78000 },
  { bike_name: 'Royal Enfield Classic 350', price: 210000 }
]
vehicles> db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
[
  { vehicle_name: 'Maruti Suzuki Swift', price: 650000 },
  { vehicle_name: 'Tata Nexon', price: 850000 },
  { vehicle_name: 'Mahindra Bolero', price: 950000 },
  { vehicle_name: 'Ashok Leyland Dost', price: 780000 },
  { vehicle_name: 'Tata Intra', price: 920000 }
]
vehicles> |
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
vehicles> db.two_wheelers.find({ manufacturer: "Hero MotoCorp" }).pretty()
[
  {
    _id: ObjectId('699fd27d786c560db87c2907'),
    bike_name: 'Hero Splendor Plus',
    model: 'gearless',
    category: '100cc',
    colors_available: [ 'Red', 'Black', 'Blue' ],
    manufacturer: 'Hero MotoCorp',
    performance: 7.5,
    timestamp: ISODate('2023-03-15T00:00:00.000Z'),
    price: 72000
  }
]
vehicles> |
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
vehicles> db.two_wheelers.find({ manufacturer: "Hero MotoCorp" }).pretty()
[
  {
    _id: ObjectId('699fd27d786c560db87c2907'),
    bike_name: 'Hero Splendor Plus',
    model: 'gearless',
    category: '100cc',
    colors_available: [ 'Red', 'Black', 'Blue' ],
    manufacturer: 'Hero MotoCorp',
    performance: 7.5,
    timestamp: ISODate('2023-03-15T00:00:00.000Z'),
    price: 72000
  }
]
vehicles> db.four_wheelers.find({ variants: "Diesel" }).pretty()
[
  {
    _id: ObjectId('699fd2c3786c560db87c290c'),
    vehicle_name: 'Maruti Suzuki Swift',
    model: 'own',
    category: 'car',
    variants: [ 'VXi', 'ZXi', 'Petrol', 'Diesel' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8.5,
    timestamp: ISODate('2023-06-10T00:00:00.000Z'),
    price: 650000
  },
  {
    _id: ObjectId('699fd2c3786c560db87c290d'),
    vehicle_name: 'Tata Nexon',
    model: 'own',
    category: 'car',
    variants: [ 'VXi', 'ZXi', 'Petrol', 'Diesel' ],
    manufacturer: 'Tata Motors',
    performance: 8.5,
    timestamp: ISODate('2023-06-10T00:00:00.000Z'),
    price: 850000
  }
]
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

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```
mongosh mongodb://127.0.0.1:27026
vehicles> db.two_wheelers.find(
  { performance: { $gt: 5 } },
  { bike_name: 1, category: 1, manufacturer: 1, _id: 0 }
).pretty()
[
  {
    bike_name: 'Hero Splendor Plus',
    category: '100cc',
    manufacturer: 'Hero MotoCorp'
  },
  {
    bike_name: 'Bajaj Pulsar NS200',
    category: '200cc',
    manufacturer: 'Bajaj Auto'
  },
  {
    bike_name: 'TVS Apache RTR 160',
    category: '160cc',
    manufacturer: 'TVS Motor'
  },
  {
    bike_name: 'Honda Activa 6G',
    category: '110cc',
    manufacturer: 'Honda'
  },
  {
    bike_name: 'Royal Enfield Classic 350',
    category: '350cc',
    manufacturer: 'Royal Enfield'
  }
]
vehicles> |
```

```
mongosh mongodb://127.0.0.1:27026
vehicles> db.four_wheelers.find(
  { performance: { $gt: 5 } },
  { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 }
).pretty()
[
  {
    vehicle_name: 'Maruti Suzuki Swift',
    category: 'car',
    manufacturer: 'Maruti Suzuki'
  },
  {
    vehicle_name: 'Tata Nexon',
    category: 'car',
    manufacturer: 'Tata Motors'
  },
  {
    vehicle_name: 'Mahindra Bolero',
    category: 'mini truck',
    manufacturer: 'Mahindra'
  },
  {
    vehicle_name: 'Ashok Leyland Dost',
    category: 'mini truck',
    manufacturer: 'Ashok Leyland'
  },
  {
    vehicle_name: 'Tata Intra',
    category: 'lorry',
    manufacturer: 'Tata Motors'
  }
]
vehicles>
vehicles>
vehicles>
vehicles>
vehicles> |
```

2. Use MongoDB to implement the following DB operations for a Zoo

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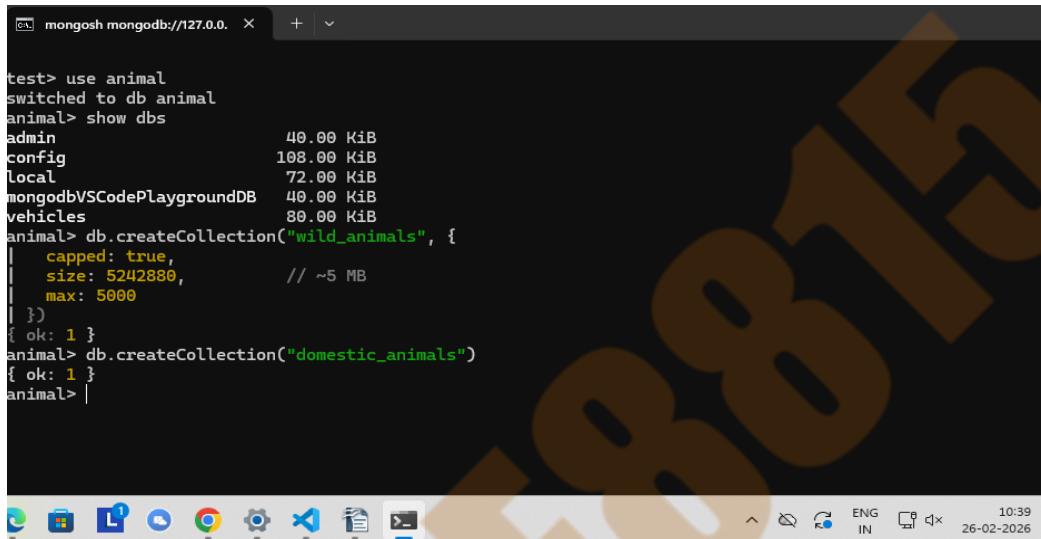
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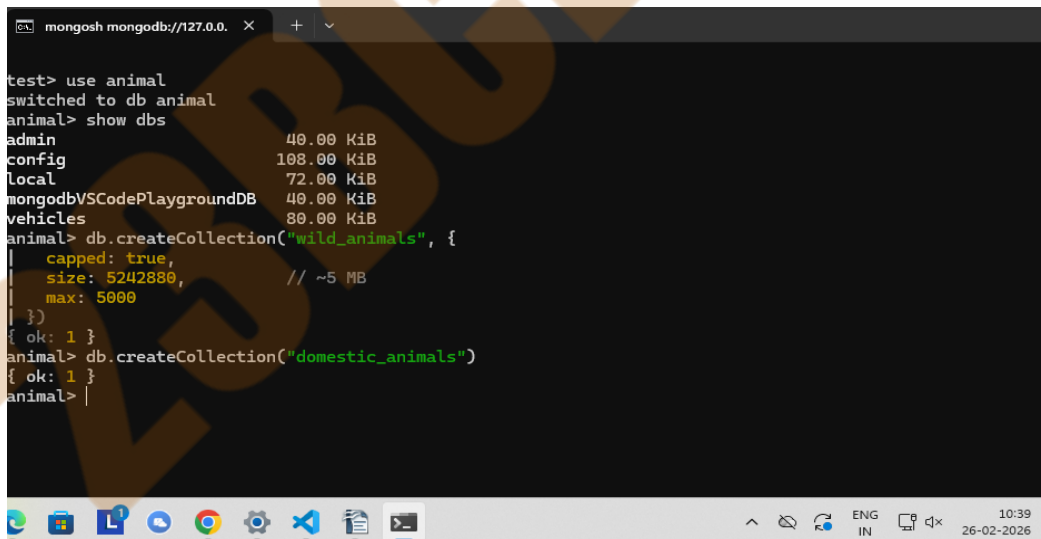
Reg. no.:

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



```
mongosh mongodb://127.0.0.1:27017
test> use animal
switched to db animal
animal> show dbs
admin                40.00 KiB
config               108.00 KiB
local                72.00 KiB
mongodbVSCodePlaygroundDB 40.00 KiB
vehicles             80.00 KiB
animal> db.createCollection("wild_animals", {
  capped: true,
  size: 5242880, // ~5 MB
  max: 5000
})
{ ok: 1 }
animal> db.createCollection("domestic_animals")
{ ok: 1 }
animal>
```

2. Write a MongoDB query to display all the databases.



```
mongosh mongodb://127.0.0.1:27017
test> use animal
switched to db animal
animal> show dbs
admin                40.00 KiB
config               108.00 KiB
local                72.00 KiB
mongodbVSCodePlaygroundDB 40.00 KiB
vehicles             80.00 KiB
animal> db.createCollection("wild_animals", {
  capped: true,
  size: 5242880, // ~5 MB
  max: 5000
})
{ ok: 1 }
animal> db.createCollection("domestic_animals")
{ ok: 1 }
animal>
```

3. Create a collection called 'wild_animals' (use capping) and Create a collection called 'domestic_animals'.
4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

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```
mongosh mongodb://127.0.0.1:27020/
> use zoo
> insertMany([
  {
    favorite_foods: ["grass", "banana", "sugarcane"],
    care_taker_name: "Suresh Patil",
    life_span: 65,
    timestamp: ISODate("2022-11-03T09:15:00Z"),
    expenses: 420000
  },
  {
    animal_name: "Indian Leopard",
    nature: "harm",
    favorite_foods: ["deer", "monkey", "rabbit"],
    care_taker_name: "Ravi Kumar",
    life_span: 15,
    timestamp: ISODate("2024-01-20T14:45:00Z"),
    expenses: 195000
  },
  {
    animal_name: "Sloth Bear",
    nature: "harm",
    favorite_foods: ["honey", "termites", "fruits"],
    care_taker_name: "Anita Sharma",
    life_span: 20,
    timestamp: ISODate("2023-08-10T11:20:00Z"),
    expenses: 165000
  },
  {
    animal_name: "Indian Gaur",
    nature: "harmless",
    favorite_foods: ["grass", "leaves", "bamboo"],
    care_taker_name: "Suresh Patil",
    life_span: 20,
    timestamp: ISODate("2021-12-05T08:00:00Z"),
    expenses: 180000
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd5ec809e0f61cb7c2907'),
    '1': ObjectId('699fd5ec809e0f61cb7c2908'),
    '2': ObjectId('699fd5ec809e0f61cb7c2909'),
    '3': ObjectId('699fd5ec809e0f61cb7c290a'),
    '4': ObjectId('699fd5ec809e0f61cb7c290b')
  }
}
animal>
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

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```
mongosh mongodb://127.0.0.1:27017/
> use wild_animals
wild_animals> find()
{
  favorite_foods: ["chicken", "rice", "pedigree"],
  animal_petname: "Rocky",
  life_span: 13,
  timestamp: ISODate("2024-02-10T15:30:00Z"),
  expenses: 32000
},
{
  animal_name: "Goat",
  gender: "female",
  favorite_foods: ["leaves", "grass", "kitchen waste"],
  animal_petname: "Meena",
  life_span: 12,
  timestamp: ISODate("2022-09-18T10:10:00Z"),
  expenses: 28000
},
{
  animal_name: "Cat",
  gender: "female",
  favorite_foods: ["fish", "milk", "whiskas"],
  animal_petname: "Misty",
  life_span: 15,
  timestamp: ISODate("2023-11-05T18:20:00Z"),
  expenses: 38000
},
{
  animal_name: "Horse",
  gender: "male",
  favorite_foods: ["grass", "hay", "oats"],
  animal_petname: "Thunder",
  life_span: 30,
  timestamp: ISODate("2021-06-30T09:00:00Z"),
  expenses: 95000
}
}
}
> use domestic_animals
domestic_animals> find()
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd647809e0f61cb7c290c'),
    '1': ObjectId('699fd647809e0f61cb7c290d'),
    '2': ObjectId('699fd647809e0f61cb7c290e'),
    '3': ObjectId('699fd647809e0f61cb7c290f'),
    '4': ObjectId('699fd647809e0f61cb7c2910')
  }
}
> animal>
```

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

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```
mongosh mongodb://127.0.0.1:27020
> use animal
> db.domestic_animals.find().pretty()
{
  "_id": ObjectId("699fd647809e0f61cb7c290c"),
  "animal_name": "Cow",
  "gender": "female",
  "favorite_foods": [ "grass", "hay", "cattle feed" ],
  "animal_petname": "Lakshmi",
  "life_span": 18,
  "timestamp": ISODate("2023-03-22T07:45:00.000Z"),
  "expenses": 45000
},
{
  "_id": ObjectId("699fd647809e0f61cb7c290d"),
  "animal_name": "Dog",
  "gender": "male",
  "favorite_foods": [ "chicken", "rice", "pedigree" ],
  "animal_petname": "Rocky",
  "life_span": 13,
  "timestamp": ISODate("2024-02-10T15:30:00.000Z"),
  "expenses": 32000
},
{
  "_id": ObjectId("699fd647809e0f61cb7c290e"),
  "animal_name": "Goat",
  "gender": "female",
  "favorite_foods": [ "leaves", "grass", "kitchen waste" ],
  "animal_petname": "Meena",
  "life_span": 12,
  "timestamp": ISODate("2022-09-18T10:10:00.000Z"),
  "expenses": 28000
},
{
  "_id": ObjectId("699fd647809e0f61cb7c290f"),
  "animal_name": "Cat",
  "gender": "female",
  "favorite_foods": [ "fish", "milk", "whiskas" ],
  "animal_petname": "Misty",
  "life_span": 15,
  "timestamp": ISODate("2023-11-05T18:20:00.000Z"),
  "expenses": 38000
},
{
  "_id": ObjectId("699fd647809e0f61cb7c2910"),
  "animal_name": "Horse",
  "gender": "male",
  "favorite_foods": [ "grass", "hay", "oats" ],
  "animal_petname": "Thunder",
  "life_span": 30,
  "timestamp": ISODate("2021-06-30T09:00:00.000Z"),
  "expenses": 95000
}
]
animal> |
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
animal> // Wild animals - name & expenses only
> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
{
  "animal_name": "Cow",
  "expenses": 45000
},
{
  "animal_name": "Dog",
  "expenses": 32000
},
{
  "animal_name": "Goat",
  "expenses": 28000
},
{
  "animal_name": "Cat",
  "expenses": 38000
},
{
  "animal_name": "Horse",
  "expenses": 95000
}
]
animal> |
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

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```
animal> db.domestic_animals.find({ life_span: 15 }).pretty()
[
  {
    _id: ObjectId('699fd647809e0f61cb7c290f'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk', 'whiskas' ],
    animal_petname: 'Misty',
    life_span: 15,
    timestamp: ISODate('2023-11-05T18:20:00.000Z'),
    expenses: 38000
  }
]
animal> |
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

```
animal> db.wild_animals.find({ care_taker_name: "Ravi Kumar" }).pretty()
[
  {
    _id: ObjectId('699fd5ec809e0f61cb7c2907'),
    animal_name: 'Bengal Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer', 'wild boar' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 12,
    timestamp: ISODate('2023-05-14T10:30:00.000Z'),
    expenses: 285000
  },
  {
    _id: ObjectId('699fd5ec809e0f61cb7c2909'),
    animal_name: 'Indian Leopard',
    nature: 'harm',
    favorite_foods: [ 'deer', 'monkey', 'rabbit' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 15,
    timestamp: ISODate('2024-01-20T14:45:00.000Z'),
    expenses: 195000
  }
]
animal> |
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
animal>
|
| db.wild_animals.find(
|   { life_span: { $gt: 5 } },
|   { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
| ).pretty()
|
| db.domestic_animals.find(
|   { life_span: { $gt: 5 } },
|   { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
| ).pretty()
| [
|   {
|     animal_name: 'Cow',
|     favorite_foods: [ 'grass', 'hay', 'cattle feed' ],
|     expenses: 45000
|   },
|   {
|     animal_name: 'Dog',
|     favorite_foods: [ 'chicken', 'rice', 'pedigree' ],
|     expenses: 32000
|   },
|   {
|     animal_name: 'Goat',
|     favorite_foods: [ 'leaves', 'grass', 'kitchen waste' ],
|     expenses: 28000
|   },
|   {
|     animal_name: 'Cat',
|     favorite_foods: [ 'fish', 'milk', 'whiskas' ],
|     expenses: 38000
|   },
|   {
|     animal_name: 'Horse',
|     favorite_foods: [ 'grass', 'hay', 'oats' ],
|     expenses: 95000
|   }
| ]
animal> |
```